Arham Ashfaq

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Profile

Power electronics engineer working on wide-bandgap hardware. I completed my MSc in Electronics Engineering at SDU in October 2026, with the highest grade for a thesis on GaN switching behaviour. In research with Danfoss, I designed a GaN test board with isolated gate drive and auxiliary supplies, used Ansys Q3D to control power-loop parasitics at 800 V, and ran more than 100 switching tests. I have designed an EMC input filter verified against EN 55011 and isolated and non-isolated converters through to bench validation. Two years in industry added FMEA, root cause analysis, and safety-critical design to SIL requirements.

Professional Experience

Student Research Assistant *Oct 2024 – Present*
University of Southern Denmark (SDU), Danfoss collaboration, Power Electronics R&D Sønderborg, Denmark

- Designed the schematic of a GaN Double Pulse Test board from device datasheets, selecting gate isolators, isolated auxiliary supplies, and slew-rate resistors, and laid out the board in Altium Designer.
- Extracted power-loop parasitic inductance and capacitance in Ansys Q3D and fed the results into layout changes to reduce overshoot at around 100 V/ns on an 800 V DC link; checked the models with impedance analyzer measurements of the built boards.
- Ran 100+ Double Pulse Tests on 30 mΩ and 50 mΩ GaN devices across 200–800 V and 2–15 A under a supervisor; wrote MATLAB scripts for switching-energy extraction, curve fitting, and statistical analysis.
- Built PLECS loss and thermal models from measured parameters; compared NPC and flying-capacitor inverters, GaN against Si, across load and temperature, including the effect of long motor cables on overshoot; prepared reports presenting the results.

Instrument & Control Engineer *Jun 2023 – Aug 2024*
Fatima Fertilizer Company Limited Sadiqabad, Pakistan

- Designed safety logic to SIL hardware fault tolerance and led the move from 1oo1 to 2oo2/2oo3 voting, removing single points of failure.
- Ran root cause analysis (5-Why, FMEA) on equipment failures and proposed corrective actions; brought equipment into service through FAT and SAT; wrote specifications and evaluated supplier offers against them.
- Authorised Permit to Work issuer for electrical isolation; led 10 technicians and two engineers across three plants and presented technical reports to management.

Graduate Trainee Engineer *Jun 2022 – Jun 2023*
Fatima Fertilizer Company Limited, Electrical & Instrumentation Sadiqabad, Pakistan

- Commissioned two 25 kVA UPS units and two 10 kVA battery chargers, tracing a recurring UPS fault to its cause; trained on MV/LV switchgear, protection relays, and VFDs.

Projects

EMC Filter for a 60 W Modulated-Frequency Boost Converter (hardware lead) *Sep – Dec 2025*

- Designed the CM/DM input filter against Middlebrook stability criteria, selected and wound the chokes, and verified conducted emissions against EN 55011 iteratively on a Rohde & Schwarz EMI receiver.

110 W Dual-Output Synchronous Buck Converter (hardware lead in a team) *Sep 2025 – Jan 2026*

- Took the design from specification and LTspice simulation through gate-drive and inductor design and Altium layout to bench verification at 97% peak efficiency.

200 W Current-Mode Flyback Converter (hardware lead in a team) *Sep 2024 – Jan 2025*

- Designed the isolating HF transformer, RCD snubber, and Type-II compensation; verified loop margins in LTspice and measured efficiency, switching stress, and thermal performance.

LLC Resonant Converter with Microcontroller Control (hardware lead in a team) *Feb – Jun 2025*

- Designed the resonant tank and voltage and current sensing for closed-loop control, and laid out the board in Altium. It held 12 V at no load but sagged under load, pointing to the control loop, which stayed unresolved at the deadline.

Publications

First author: GaN switching loss characterization in a flying-capacitor converter using Double Pulse Testing (submitted, under review). A second paper on snubber optimization for the same test setup is in preparation.

Education

MSc, Electronics Engineering *2024 – 2026*
University of Southern Denmark (SDU) Sønderborg, Denmark

- Completed October 2026. Thesis (highest grade): Double Pulse Test Based Experimental Analysis of GaN Switching Behavior for High-Performance Motor Drive Design.

BSc, Electrical & Electronics Engineering (CGPA: 3.43/4.00) *2018 – 2022*
University of Engineering & Technology (UET), Lahore Lahore, Pakistan

- Final-year project: Three-Phase Grid-Tied Solar Inverter with Resonant Converter.

Technical Skills

Power Hardware	GaN and Si MOSFETs, isolated gate drive, auxiliary supplies, snubbers, magnetics, thermal design; buck, flyback, LLC, NPC, FLC
EMI/EMC	CM/DM filter design, Middlebrook stability, EN 55011 conducted emissions, power-loop layout
PCB & Simulation	Altium Designer, KiCad; Ansys Q3D, LTspice, PSpice, PLECS, MATLAB/Simulink
Embedded & Control	TI C2000, Xilinx Zynq-7000, Arduino; C, C++, Python
Test & Reliability	6 GHz oscilloscopes, HV probes, impedance analyzer, R&S EMI receiver; FMEA, 5-Why, SIL

Languages

English (C1) | **Urdu** (Native) | **Danish** (Learning)

Additional Relevant Information

Holds a Danish Job Search visa valid until 2029, with a long-term plan to build a career in Denmark. Willing to relocate anywhere in Denmark. Available to start immediately.